

FORMULARIO-MATRICIAL-RETICULADAS...



user_3135346



Ingenieria Estructural



1º Máster en Ingeniería Industrial



**Escuela Politécnica Superior. Campus de Leganés
Universidad Carlos III de Madrid**



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antes



**Descarga sin publi
con 1 coin**



Después

WUOLAH



Importante

Puedo eliminar la publi de este documento con 1 coin

¿Cómo consigo coins?

Plan Turbo: barato

Planes pro: más coins

FÓRMULAS MATRICIAL RETICULADAS

$$[K'] = \begin{bmatrix} (K_{11}) & (K_{12}) \\ (K_{21}) & (K_{22}) \end{bmatrix}$$

$$[K] = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix}$$

$$[T] = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[T]^T = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[K]^6 = [T] [K]^L [T]^T$$

$$[K]^6 = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A & 0 & 0 \\ 0 & B & C \\ 0 & D & E \end{bmatrix} \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\cos 0 = 1$$

$$\cos 90 = 0$$

$$\sin 0 = 0$$

$$\sin 90 = 1$$

$$\cos 180 = -1$$

$$\cos 270 = 0$$

$$\sin 180 = 0$$

$$\sin 270 = -1$$

$$\cos 45 = \frac{\sqrt{2}}{2}$$

$$\cos 225 = -\frac{\sqrt{2}}{2}$$

$$\sin 45 = \frac{\sqrt{2}}{2}$$

$$\sin 225 = -\frac{\sqrt{2}}{2}$$

$$\cos 135 = -\frac{\sqrt{2}}{2}$$

$$\cos 315 = \frac{\sqrt{2}}{2}$$

$$\sin 135 = \frac{\sqrt{2}}{2}$$

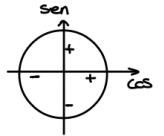
$$\sin 315 = -\frac{\sqrt{2}}{2}$$

$$\cos 60 = \frac{1}{2}$$

$$\cos 30 = \frac{\sqrt{3}}{2}$$

$$\sin 60 = \frac{\sqrt{3}}{2}$$

$$\sin 30 = \frac{1}{2}$$



MATRIZ DE RIGIDEZ GLOBAL

$$(K_{11})^6 = \begin{bmatrix} A \cos^2\alpha + B \sin^2\alpha & -A \cos\alpha \sin\alpha + B \cos\alpha \sin\alpha & C \sin\alpha \\ -A \cos\alpha \sin\alpha + B \cos\alpha \sin\alpha & A \sin^2\alpha + B \cos^2\alpha & C \cos\alpha \\ D \sin\alpha & D \cos\alpha & E \end{bmatrix}$$

$$(K_{22})^6 = \begin{bmatrix} A \cos^2\alpha + B \sin^2\alpha & -A \cos\alpha \sin\alpha + B \cos\alpha \sin\alpha & -C \sin\alpha \\ -A \cos\alpha \sin\alpha + B \cos\alpha \sin\alpha & A \sin^2\alpha + B \cos^2\alpha & -C \cos\alpha \\ -D \sin\alpha & -D \cos\alpha & E \end{bmatrix}$$

$$(K_{12})^6 = \begin{bmatrix} A \cos^2\alpha - B \sin^2\alpha & -A \cos\alpha \sin\alpha - B \cos\alpha \sin\alpha & C \sin\alpha \\ -A \cos\alpha \sin\alpha - B \cos\alpha \sin\alpha & A \sin^2\alpha - B \cos^2\alpha & C \cos\alpha \\ -D \sin\alpha & -D \cos\alpha & E \end{bmatrix}$$

$$(K_{21})^6 = \begin{bmatrix} A \cos^2\alpha + B \sin^2\alpha & -A \cos\alpha \sin\alpha - B \cos\alpha \sin\alpha & -C \sin\alpha \\ -A \cos\alpha \sin\alpha - B \cos\alpha \sin\alpha & A \sin^2\alpha - B \cos^2\alpha & -C \cos\alpha \\ D \sin\alpha & D \cos\alpha & E \end{bmatrix}$$

ESFUERZOS EN BARRAS EN EJES LOCALES

$$\{s\}' = [K]^{-1} [T]^T \{d\}' + \{r\}'$$

en la barra

$$\begin{bmatrix} S_{x1} \\ S_{y1} \\ S_{\theta 1} \\ S_{x2} \\ S_{y2} \\ S_{\theta 2} \end{bmatrix}' = \begin{bmatrix} N_1 \\ Q_1 \\ M_1 \\ N_2 \\ Q_2 \\ M_2 \end{bmatrix}' = \begin{bmatrix} (K_{11})^{12} & (K_{12})^{12} \\ (K_{21})^{12} & (K_{22})^{12} \end{bmatrix}' [T]^T \begin{bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{bmatrix} + [T]^T \begin{bmatrix} r_{M1} \\ r_{Q1} \\ r_{M2} \\ r_{Q2} \\ r_{M2} \end{bmatrix}'$$

NOBOA

$$\begin{bmatrix} N_1 \\ Q_1 \\ M_1 \end{bmatrix}' = [(K_{11})^{12}]^L [T]^T \begin{bmatrix} u_1 \\ v_1 \\ \theta_1 \end{bmatrix} + [(K_{12})^{12}]^L [T]^T \begin{bmatrix} u_2 \\ v_2 \\ \theta_2 \end{bmatrix} + [T]^T \begin{bmatrix} r_{M1} \\ r_{Q1} \\ r_{M1} \end{bmatrix}'$$

NOBOA 2

$$\begin{bmatrix} N_2 \\ Q_2 \\ M_2 \end{bmatrix}' = [(K_{21})^{12}]^L [T]^T \begin{bmatrix} u_1 \\ v_1 \\ \theta_1 \end{bmatrix} + [(K_{22})^{12}]^L [T]^T \begin{bmatrix} u_2 \\ v_2 \\ \theta_2 \end{bmatrix} + [T]^T \begin{bmatrix} r_{N2} \\ r_{Q2} \\ r_{M2} \end{bmatrix}'$$

perdo espacio



Necesito concentración

ali ali oohh
esto con 1 coin me
lo quito yo...

WUOLAH

WUOLAH